Spot size Limitation

Monday, March 24, 2025 3:03 PM

Theoretical Calculation:

$\lambda = \frac{v}{f}$ Spot size $\approx 2.5\ \mu m \pm 0.5\ \mu m$

Parameters: AlScN / Si

Case 1: $f = 5\ GHz = 5\times10^9\ Hz$

$v = 4.883\times10^3\ m/s$ (Sezawa mode)

$\lambda = \frac{4.883\times10^3}{5\times10^9}$

$= 9.77\times10^{-7}\ m$

$= 0.98\ \mu m$

$\lambda = 980\ nm$

Case 2: $f = 10\ GHz = 10\times10^9\ Hz$

$v = 4.883\times10^3\ m/s$ (Sezawa mode)

$\lambda = \frac{4.883\times10^3}{10\times10^9}$

$= 4.88\times10^{-7}\ m$

$= 0.488\ \mu m$

$\lambda = 488\ nm$

Case 1: $\frac{2.5\ \mu m}{0.98\ \mu m} = 2.55$ [Spot size is 2-55 times λ]

Case 2: $\frac{2.5\ \mu m}{0.49\ \mu m} = 5.10$ [Spot size is 5-10 times λ]

When modeling the optical measurement as a Spatial Averaging

The Spot size is a convolution operator

Field: (Acoustic) $E(x) = E_0 e^{i\phi(x)}$

Probe intensity profile (optical): $G(x) = \frac{1}{\sqrt{\pi}\,\omega_0} e^{\left(-x^2/\omega_0^2\right)}$ Gaussian

$\omega_0 = 2.5\ \mu m$

Phase modulation of optical signal Sin Mod.

$\phi(x) = A\cos(k_a x)$

$k_a = 2\pi/\lambda$

The measured field (optical) is the convolution of

$E(x)$ (Acoustic) with $G(x)$ (Optical Gaussian)

High spatial frequency gets filtered

Fourier transform of Gaussian is Gaussian

$\tilde{G}(k) = \exp\left(-\frac{(\omega_0 k)^2}{4}\right)$

k (Spatial freq. component) is attenuated by

$H(k) = \exp\left(-\frac{(\omega_0 k)^2}{4}\right)$

$H(k_a) = \exp\left[-\frac{(\omega_0 k_a)^2}{4}\right]$ $k_a = 2\pi/\lambda$

$H(k_a) = \exp\left[-\frac{\omega_0^2}{4}\left(\frac{2\pi}{\lambda}\right)^2\right]$

Now,

Case 1:

$k_a = \frac{2\pi}{0.98\ \mu m}$

$k_a = 6.41\ \mu m^{-1}$

$H(k_a) = \exp\left[-\frac{(2.5\times10^6\times6.41\times10^6)^2}{4}\right]$

$= \exp\left[-\frac{(16.03)^2}{4}\right]$

$= \exp\left[-\frac{257}{4}\right]$

$= \exp(-64.25)$

$H(k_a) = 1.24\times10^{-28} \ll 0$

Case 2:

$k_a = \frac{2\pi}{0.49\ \mu m}$

$k_a = 12.82\ \mu m^{-1}$

$H(k_a) = \exp\left[-\frac{(2.5\times10^6\times12.82\times10^6)^2}{4}\right]$

$= \exp\left[-\frac{(32.05)^2}{4}\right]$

$= \exp\left[-\frac{1029}{4}\right]$

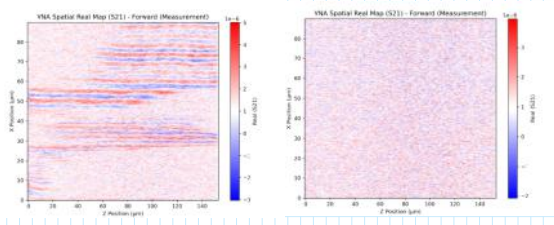
$= e^{-256.98}$

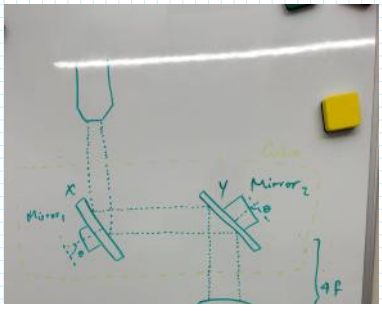
$H(k_a) = 3.12\times10^{-112} \ll 0$

Spot $= 2.5 \pm 0.5\ \mu m$ Spot $\approx 2.5 \pm 0.5\ \mu m$

$\lambda \approx 4.4\ \mu m$ $\lambda \approx 1.46\ \mu m$

Y-cut LN/Si split IDT @ 1 GHz @3.1 GHz





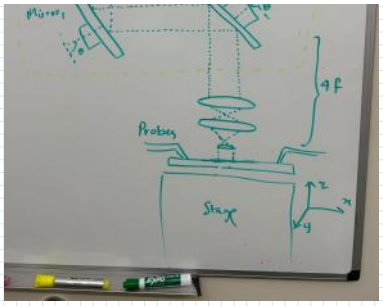
- Wire Bonding routes for pads?
- Visible wavelength?

11.17 mm → Perfectly Balanced

$= \exp\left[-\frac{10^{29}}{\Delta}\right]$

$= e^{-256.78}$

$H(k_x) = 3.12 \times 10^{-112} <<< 0$



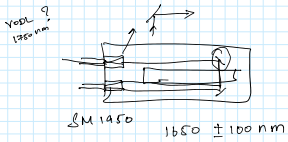
The spot size of the LTF attenuates high spatially varying frequencies $> 1/w_0$

LTF is then like a low-pass filter where it can't resolve with high resolution and averages the variation (phase modulation)

$\therefore$ As we scan along the wavefronts we can capture the wavfronts and can apply time evolution animation by varying the $Re(S_z)\, e^{i\phi}$, $\phi = 0 \to 2\pi$.

11.17 mm → Perfectly Balanced

$\frac{1}{P}\frac{dP}{df} \approx 3.6 \times 10^{-10}$ /Hz



Ideas

- Synthetic wavelength

- Structured light
- Lightsheet Microscopy
- Metalens (Bi-directional)
- Solid Immersion lens.
- 4f + Galvo
- Wire Bonding electrical pads

d = spot diameter.

for 10 GHz

$\lambda = 400$ nm

Sample $\approx \lambda/2 \approx 200$ nm (spot)

$NA = n \sin\theta$ $\quad z_R = \frac{\pi w_0^2}{\lambda}$

$\lambda_1 = 1550$ nm $\quad NA = \frac{2\lambda}{\pi d}$

$\lambda_2 = 1750$ nm

$d = \frac{2\lambda}{\pi NA}$ $\quad NA \approx 1.4$ $\quad NA = 1$

darkring of Airy disk

1550 $\left\{ \begin{array}{l} d = \frac{2(1.55\times10^{-6})}{\pi(1.4)} \\ d = 704.82\,nm \end{array} \right.$ $\quad d = 986\,nm$

1750 $\left\{ \begin{array}{l} d = \frac{2(1.75\times10^{-6})}{\pi(1.4)} \\ d = 795.775\,nm \end{array} \right.$ $\quad d = 1.114\,\mu m$

limit is $\lambda/2$ in far field.

$1550 \to \lambda/2 \approx 775$ nm
$1750 \to \lambda/2 \approx 875$ nm } limit of spot diameter in far field.

Solution?

- High NA
- Galvo + Lens System
- Visible wavelength → Ti-Sapphire 650 - 1100 nm ↓ λ/2 325 nm
- Near field ↓ Plasmonic tips & Metamaterials Sub-λ focus → 400 nm (violet) ↓ λ/2 200 nm limit

Polytec LDV

Spatial Resolution ~2 nm

Displacement ~pm

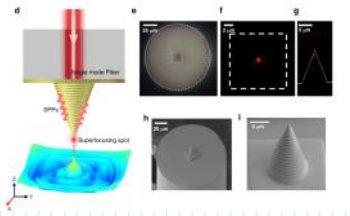
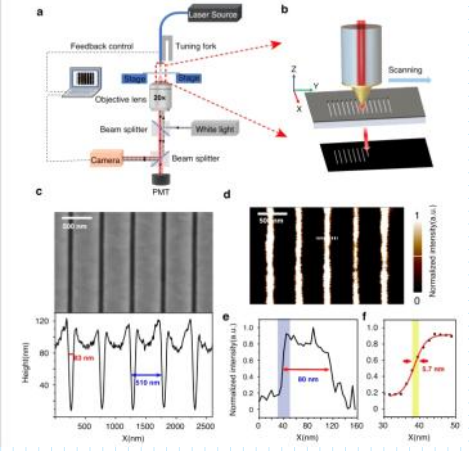


Fig. 4: Experimental characterization of home-made SNOM tip resolution by performing near-field imaging upon a standard chrome lamellar-slits grating.



Options

① 1550 nm + 2.5 → 1.4/1.5 μm Spot LTF ⇓ upto 3 GHz

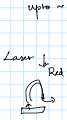
② 1550 nm + Galvo + Lens System ⇓ upto 3-4 GHz

③ 1750 nm + Galvo + Lens system ⇓ upto 2-4 GHz

④ Visible wavelength 400 nm (violet)/650 nm Ti-Sapphire + Galvo lens System → Spot 200 nm ⇓ upto ~10 GHz

Coupling tip → Cavity Probe → Size?

Laser ↓ Red



Si Cavity

Calc.
Cavity made
Sensitivity

Issue

* 1550 / 1750 nm + 2.5 ± 0.5 μm LTF ⇓ upto 1 GHz

Galvo_system_calcul...

Galvo_system_calcul...

Confocal Galvo Scanning System Calculations for 1550 nm Laser

1. Mode Field Diameter and Collimated Beam Size

$$\mathrm{MFD}_{\mathrm{SMF-28}} \approx 10.5\,\mu\mathrm{m}$$
$$w_0 = \frac{\mathrm{MFD}}{2} \approx \frac{10.5}{2} = 5.25\,\mu\mathrm{m}$$

Beam divergence angle (half-angle):

$$\theta = \frac{\lambda}{\pi w_0} = \frac{1.55 \times 10^{-6}\,\mathrm{m}}{\pi \cdot 5.25 \times 10^{-6}\,\mathrm{m}} \approx 0.094\,\mathrm{rad}$$

Collimated beam radius after fiber collimator:

Let $f_{\mathrm{col}} = 12\,\mathrm{mm}$:

$$w = f_{\mathrm{col}} \cdot \theta = 12 \times 10^{-3}\,\mathrm{m} \cdot 0.094 = 1.13\,\mathrm{mm}$$
$$\text{Beam diameter } D = 2w \approx 2.26\,\mathrm{mm}$$

2. Galvo Mirror Repeatability

Given closed-loop repeatability of the galvo system:

$$\theta_{\mathrm{rep}} = 15\,\mu\mathrm{rad}$$

If the beam is focused at a distance L from the galvo (after scanning and relay optics), then the lateral repeatability is:

$$\Delta x = L \cdot \theta_{\mathrm{rep}}$$

Example: if $L = 200\,\mathrm{mm}$:

$$\Delta x = 200 \times 10^{-3}\,\mathrm{m} \cdot 15 \times 10^{-6}\,\mathrm{rad} = 3\,\mu\mathrm{m}$$

3. 4f Relay Beam Reduction

Let:

$$f_1 = 101.6\,\mathrm{mm}$$
$$f_2 = 50.8\,\mathrm{mm}$$

Magnification:

$$M = \frac{f_2}{f_1} = \frac{50.8}{101.6} = 0.5$$

Input beam diameter:

$D_{\mathrm{in}} = 2.26\,\mathrm{mm}$

$$D_{\mathrm{out}} = M \cdot D_{\mathrm{in}} = 0.5 \cdot 2.26 = 1.13\,\mathrm{mm}$$

4. Focusing and Diffraction-Limited Spot Size

Let:

$$\lambda = 1.55\,\mu\mathrm{m}$$
$$\mathrm{NA} = 0.4$$

$$d = \frac{\lambda}{2 \cdot \mathrm{NA}} = \frac{1.55}{2 \cdot 0.4} = 1.9375\,\mu\mathrm{m}$$

For a spot size of $\lambda/2 = 0.775\,\mu\mathrm{m}$, required NA:

$$\mathrm{NA}_{\lambda/2} = \frac{\lambda}{2d} = \frac{1.55}{2 \cdot 0.775} = 1.0$$

Galvo_system_calcul...

Detailed Optical Design and Scan Repeatability Optimization

1. System Overview

The optical path includes:

1. Fiber output (SMF-28 or equivalent) at 1550 nm
2. Fiber collimator
3. 2-axis Galvo scanner
4. 4f beam reduction telescope
5. Objective lens
6. Sample

2. Collimation

Input Parameters:

- Mode Field Diameter (MFD): 10.5 µm
- Wavelength: 1550 nm
- Fiber collimator focal length: 12 mm

$$
\begin{aligned}
w_0 &= \frac{10.5}{2} = 5.25\,\mu\text{m} \\
\theta &= \frac{\lambda}{\pi w_0} = \frac{1.55 \times 10^{-6}}{\pi \cdot 5.25 \times 10^{-6}} \approx 0.094\,\text{rad} \\
w &= f_{\text{col}} \cdot \theta = 12 \times 10^{-3} \cdot 0.094 \approx 1.13\,\text{mm}
\end{aligned}
$$

3. Galvo Placement

- Recommended distance from collimator to galvo input mirror: 30 mm
- Distance between galvo mirrors (typical): 13.5 mm (as per Thorlabs GVS211)

4. 4f Relay Optics Design

- Lens 1 focal length: $f_1 = 100\,\text{mm}$
- Lens 2 focal length: $f_2 = 50\,\text{mm}$
- Separation between lenses: $f_1 + f_2 = 150\,\text{mm}$
- Aperture of each lens should exceed beam diameter by 30%:

$$\text{Min. aperture} = 1.13\,\text{mm} \cdot 1.3 \approx 1.47\,\text{mm} \Rightarrow \text{use 1 in}(25.4\,\text{mm})optics$$

5. Objective Lens and Focused Spot

- Required NA to achieve diffraction-limited spot $d = 0.775\,\mu\text{m}$:

$$\text{NA} = \frac{\lambda}{2d} = \frac{1.55}{2 \cdot 0.775} = 1.0$$

- Use high-NA (~ 1.0) objective, e.g., Mitutoyo Plan Apo NIR (NA 0.7–1.0)
- Working distance (WD): 1 mm – 3.5 mm

6. Beam Path Summary (Recommended Distances)

- Fiber collimator to Galvo input: 30 mm
- Galvo mirror spacing: 13.5 mm
- Galvo output to first lens of 4f: 100 mm
- L1 to L2 spacing: 150 mm
- L2 to objective: 50 mm (match last focal plane to back aperture)
- Objective to sample: per WD, typically 1 mm

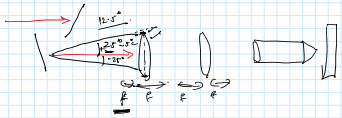
7. Improving Scan Repeatability (XY Position)

Repeatability is given by:

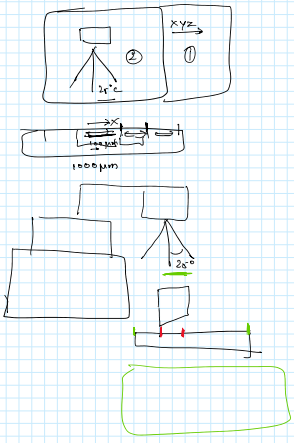
$$\Delta x = L \cdot \theta_{\text{rep}}, \quad \theta_{\text{rep}} \approx 15\,\mu\text{rad}$$

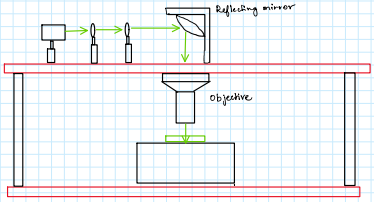
Strategies to Improve Repeatability:

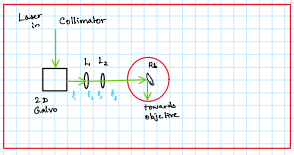
- **Minimize scan distance** L between galvo and focal plane
- **Use short focal length objective** to reduce mechanical-to-optical translation
- **High-quality galvo mirrors with better encoders**
- **Add beam stabilizer (active feedback)**
- **Temperature-stabilized environment**
- **Use pre-compensation in control signals**



15 μrad 10 mm
= 150 nm







Remember:

Galvo:
1. Wavelength range 400 - 2000 nm
   "Silver Coating"
2. Optical / mechanical Scan angle
   ± 25° / ± 12.5°
3. Repeatability of mirror
   like Step resolution
   ideal ≈ 1 μrad

4f:
1. Aperture
2. Focal length
3. Aberration free, etc,
4. Anti Reflection Coatings
5. Wavelength range.

Objective:
1. Focal length
   Working distance
2. Wavelength range and Coating
3. Magnification
   highest.

Misc.
1. Fiber Collimator
   Wavelength
2. Reflecting mirror 99.99% R
   Wavelength range.

Galvo

MEMS Mirror

4f Lens system

Objective

Galvo

MEMS Mirror

4f Lens system

Objective